ACADEMY OF TECHNOLOGY

Internal Examination (CA3), Even Semester, 2026

Programme/Discipline: Computer Science and Engineering **Semester:** 4th

Paper Name: Design and Analysis of Algorithms **Paper Code:** PCC-CS404 **UPID:** 4416

Date of Examination: 13/05/2026 **Duration:** 1 Hour **Full Marks:** 25

Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Questions No	1. i	1. ii	1. iii	1. iv	1. v	1 vi	1. vii	2	3	4	5	6	7
CO Mapping	CO1	CO1	CO2	CO3	CO3	CO2	CO3	CO4	CO4	CO2	CO1	CO4	CO3
Bloom's Level	L2	L2	L2	L3	L3	L2	L3	L4	L4	L2	L2	L4	L3

Group A

1. Attempt any five questions from the following. $[1 \times 5 = 5]$

i. Which of the following is not $O(n^2)$?

a) $(15^{10}) \times n + 12099$ b) $n^{1.98}$

c) $n^3/(\sqrt{(n)})$ d) $(2^{20}) \times n$

ii. What is the result of the recurrences which fall under second case of Master's theorem (let the recurrence be given by $T(n) = aT(n/b) + f(n)$ and $f(n) = n^c$?

a) $T(n) = O(nlog_b a)$ b) $T(n) = O(n^c log n)$

c) $T(n) = O(f(n))$ d) $T(n) = O(n^2)$

iii. Most efficient time complexity of DFS is?

a) $O(|V|)$ b) $O(|E|)$

c) $O(|V| + |E|)$ d) $O(|V| \times |E|)$

iv. Apply Quick sort on a given sequence 7 11 14 6 9 4 3 12. What is the sequence after first phase, pivot is first element?

a) 6 4 3 7 11 9 14 12 b) 6 3 4 7 9 14 11 12

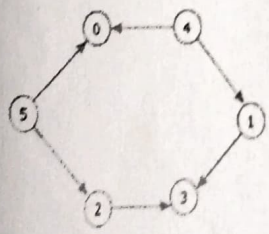
c) 7 6 14 11 9 4 3 12 d) 7 6 4 3 9 14 11 12

v. Given items as $\{profit, weight\}$ pairs $\{\{40,20\},\{30,10\},\{20,5\}\}$. The capacity of knapsack=20. Find the maximum value output assuming items to be divisible.

a) 60 b) 80

c) 100 d) 40

vi. Which of the following is a valid topological ordering of the given graph?



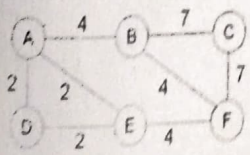
a) $5 \to 0 \to 4 \to 1 \to 2 \to 3$

b) $5 \to 2 \to 3 \to 1 \to 4 \to 0$

c) $0 \to 5 \to 2 \to 3 \to 1 \to 4$

d) $5 \to 4 \to 0 \to 1 \to 2 \to 3$

vii. Consider the given graph. What is the weight of the MST using Kruskal's algorithm?



a) 21 b) 19

c) 17 d) 20

Group B

Attempt any four questions from the following.

2. Solve the following recurrence relation using recursion tree method. [5]

$$T(n) = \begin{cases} 1 & \text{if } n = 1 \\ 2T(\frac{n}{2}) + n^2 & \text{if } n > 1 \end{cases}$$

3. Write the Algorithm for MERGE SORT, including the MERGE procedure. [5]

4. Prove that if $f(n)$ is $\Theta(g(n))$ then $g(n)$ is $\Theta(f(n))$. [5]

5. Write the BREADTH-FIRST-SEARCH ALGORITHM. Analyse the time complexity for it. [5]

6. Solve the following recurrence relation using recursion tree method. [5]

$$T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(\frac{n}{3}) + T(\frac{2n}{3}) + n & \text{if } n > 1 \end{cases}$$

7. Consider 10 jobs for A to J. The execution of each job takes 1 unit of time. One job is executed at a time. Each job is related with profit and a deadline. If job is completed before deadline then the corresponding profit is earned. Find the maximum profit. (Explain each step of your solution) [5]

Job	A	B	C	D	E	F	G	H	I	J
Profit	25	35	40	30	18	25	33	43	19	25
Deadline	4	7	4	7	1	4	6	5	3	6

ACADEMY OF TECHNOLOGY

Internal Examination (CA3), Even Sem, 2026

Programme/Discipline: **Computer Science and Engineering** Semester: **4th**
Paper Name: **Computer Architecture** Paper Code: **PCC-CS402** UPID: **4442**
Date of Examination: **12/05/2026** Duration: **1 Hour** Full Marks: **25**

Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Questions No	1. i	1. ii	1. iii	1. iv	1. v	1. vi	1. vii	2	3	4	5	6	7
CO Mapping	CO2	CO2	CO2	CO1	CO2	CO2	CO1	CO2	CO2	CO1	CO1	CO2	CO1
Bloom's Level	L1	L1	L1	L1	L1	L1	L3	L3	L2	L5	L3	L5	L4

Group A

1. Attempt any five questions from the following. [1 × 5 = 5]

i) What are the hazards in the pipeline?

ii) Utilisation pattern of successive stages of a synchronous pipeline can be specified by which table?

iii) What is forbidden latency?

iv) What are used to hold the results of intermediate stages in the pipelining?

v) What is the full form of MAL?

vi) What is the full form of ICV?

vii) A four-stage pipeline has the stage delays as 150, 120, 160 and 140 ns respectively. Registers are used between the stages and have a delay of 5 ns each. Assuming a constant clocking rate, what will be the total time required to process 1000 data items on the pipeline?

Group B

(Attempt any four questions from the following.)

[4 × 5 = 20]

2. An instruction pipeline consists of 4 stages – Fetch (F), Decode (D), Execute (E) and Write Back (W). The five instructions in a certain instruction sequence need these stages for the different number of clock cycles as shown in the table. Find the total number of clock cycles needed to perform these five instructions. [5]

Instruction	F	D	E	W
1	1	2	1	1
2	1	2	2	1
3	2	1	3	2
4	1	3	2	1
5	1	2	1	3

3. What is a branch hazard? Briefly discuss two methods to handle branch hazards. [2+3]

4. Define pipelining. Show that maximum speed-up of a pipeline is equal to number of stages. [2+3]

5. Suppose the time delays of the 4 stages of a pipeline are t1 = 60 ns, t2 = 50 ns, t3 = 90 ns and t4 = 80 ns respectively, number of instruction n = 100, and the interface latch has a delay t1 = 10 ns, then [2+3]

i) What would be the maximum clock frequency of the above pipeline?

ii) What is the speed-up of this pipeline over that of its non-pipeline counterpart?

6. Identify the hazards in the following instruction stream. [5]

I1 : r2← r5/r8

I2 : r9←r2/r7

I3 : r11←r9*r5

I4 : r10←r12

I5 : r8←r15 OR r2

7. Differentiate between static and dynamic pipeline. [5]

ACADEMY OF TECHNOLOGY
Internal Examination (CA3), Even Semester, 2026

Programme/Discipline: Computer Science and Engineering | **Semester:** 4th
Paper Name: Discrete Mathematics | **Paper Code:** PCC-CS401 | **UPID:** 4407
Date of Examination: 12/05/2026 | **Duration:** 1 Hour | **Full Marks:** 25

Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Questions No	1.i	1.ii	1.iii	1.iv	1.v	1.vi	1.vii	2	3	4	5	6	7
CO Mapping	CO3	CO3	CO3	CO1	CO1	CO1	CO1	CO2	CO3	CO3	CO3	CO2	CO2
Bloom's Level	L1	L3	L1	L2	L1	L1	L2	L3	L2	L3	L2	L3	L3

Group A

1. Attempt any five questions from the following. [1 × 5 = 5]

i. Let A and B be two non-empty sets. Then, $A\Delta B =$
(a) $A \cup B$ (b) $A \cap B$ (c) $(A \cup B) \setminus (A \cap B)$ (d) $A^c \cap B^c$.

ii. Let us consider the set of integers $\mathbb{Z}$ with the equivalence relation ρ, defined as $a\rho b$ if and only if $(a+b)$ is an even number. Then
(a) $cl(4) = \{4\}$ (b) $cl(4) = \{2, 4, 6, \cdots\}$ (c) $cl(4) = \mathbb{Z}$ (d) $cl(4) = \{0, \pm 2, \pm 4, \pm 6, \cdots\}$.

iii. The set of all irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ is
(a) finite (b) countably infinite (c) uncountable (d) none of these.

iv. Which of the following is a contradiction?
(a) $p \vee q$ (b) $p \wedge \sim p$ (c) $p \to q$ (d) $\sim p \vee q$.

v. The statement $p \to q$ is False, when
(a) $p = F, q = T$ (b) $p = F, q = F$ (c) $p = T, q = F$ (d) $p = T, q = T$.

vi. Which of the following is logically equivalent to $p \to q$?
(a) $p \wedge q$ (b) $\sim p \vee q$ (c) $\sim q \to \sim p$ (d) both (b) and (c).

vii. The negation of the statement: $\forall n \in \mathbb{Z},\ n^3 > 0$ is
(a) $\forall n \in \mathbb{Z},\ n^3 < 0$ (b) $\exists n \in \mathbb{Z},\ n^3 \leq 0$ (c) $\forall n \in \mathbb{Z},\ n^3 \leq 0$ (d) $\exists n \in \mathbb{Z},\ n^3 > 0$.

Group B

Attempt any four questions from the following.

2. Using truth table, show that the formula is a Tautology: $\sim (p \wedge q) \leftrightarrow (\sim p \vee \sim q)$. [5]

3. Define partial order relation and poset. Let S be the set of all positive divisors of 30. Draw the Hasse diagram of the poset $(S, \leq)$, where '$\leq$' defines the division. [2+3]

4. Define equivalence relation. A relation $\mathcal{R}$ defined on $\mathbb{Z}$ by "$x\mathcal{R}y$ if and only if $(x - y)$ is divisible by 5 ". Check whether $\mathcal{R}$ is an equivalence relation. [1+4]

5. Define bijective mapping. Check whether the mapping $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 3x + 1$, for all $x \in \mathbb{R}$ is bijective or not? Find $f^{-1}(x)$, if possible. [1+2+2]

6. Represents the following argument symbolically and test its validity:

I. If it rains today, then we will not meet today.

II. If we do not meet today, then we will meet tomorrow.

III. $\therefore$ If it rains today, then we will meet tomorrow. [5]

7. With or without truth table, find Conjunctive Normal Form (CNF) of the following statement: [5]

$$(p \wedge \sim (q \wedge r)) \vee (p \to q).$$

ACADEMY OF TECHNOLOGY
INTERNAL EXAMINATION EVEN SEM, 2026

Programme/Discipline: CSE
Paper Name: Formal Language and Automata Theory
Paper Code: PCC-CS403
Date of Examination: 14/05/2026
Duration: 1 Hour
Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Semester: 4th
UPID: 4423
Full Marks: 25

Questions No	1.i	1.ii	1.iii	1.iv	1.v	1.vi	1.vii	2	3	4	5	6	7
CO Mapping	CO1	CO1	CO1	CO1	CO1	CO1	CO1,CO3	CO1	CO1	CO1	CO1	CO1	CO1
Bloom's Level	L1	L3	L1	L2	L2	L3	L1	L3	L3	L3	L3	L3	L2

Group A: Attempt any five questions from the following [1x5=5]

1. i. Finite Automata requires minimum number of stacks?
   a) 1 b) 0 c) 2 d) 3

   ii. Number of states required in DFA to accept all strings ends with '101' over the alphabet set {0,1}
   a) 1 b) 2 c) 3 d) 4

   iii. How many tuples are in finite state machine?
   a) 4 b) 5 c) 6 d) unlimited

   iv. Number of states required in a DFA to accept string $a^n b^n$, where n>0.
   a) 3 b) 2 c) 6 d) can't be represented.

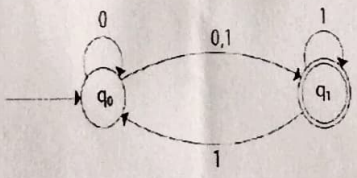
   v. What is the relation between DFA and NFA on the basis of their computational power?
   a) DFA>NFA b) NFA>DFA
   c) Equal d) can't be said

   vi. Number of states required in a DFA to accept string starts with 'aaa'.
   a) 6 b) 5
   c) 4 d) can't be represented.

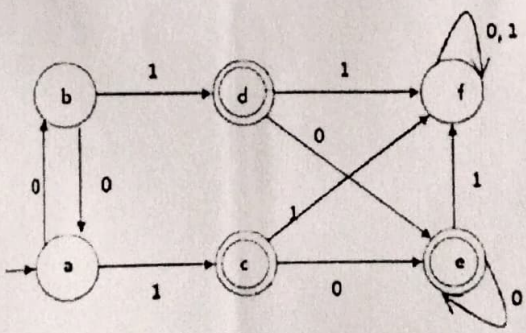
   vii. Which type of language is equivalent to the class of 'Recursively Enumerable' languages?
   a) Type 1 b) Type 0 c) Type 2 d) Type 3

Group B: Attempt any four questions from the following. [4x5=20]

2. Design a DFA over $\Sigma = \{a, b\}$ that accepts all strings containing atleast one '0' and one '1'. [5]
3. Design a DFA over $\Sigma = \{a, b\}$ that accepts all strings having even number of 'a' and odd number of 'b'. [5]
4. Convert the given NFA to DFA [5]



5. Minimize the given DFA [5]



6. Construct a DFA which accepts a language of all binary strings divisible by 3 over $\Sigma = \{0, 1\}$ [5]
7. Define trap/dead state of DFA. Explain with example. [3+2]

ACADEMY OF TECHNOLOGY
Internal Examination (CA3), Even Semester, 2026

Discipline: CSE | **Semester: 4TH**
Paper Name: Biology | **Paper Code: BSC-401**
Date of Examination: 14/05/2026 | **UPID: 4408**
Duration: 1 Hour | **Full Marks: 25**

Question-wise Mapping with Course Outcomes (COs) & Cognitive Levels Assessed:

QN	1(i)	1(ii)	1(iii)	1(iv)	1(v)	1(vi)	1(vii)	2	3	4	5	6	7
COM	CO1	C01	CO4	CO1	CO1	CO1	CO1	CO1	CO1	CO1	CO4	CO1	CO4
BL	L3	L4	L3	L1	L2	L1	L1	L2	L2	L1	L2	L1	L1

QN: Question Number **COM: Course Outcome Mapping** **BL: Bloom's Level**

Group A

1. Attempt any 5 questions from the following, choose the correct answer from the options given: **[1X5=5]**

i. **The shutter in a camera is equivalent to which part of the eye?**
a. Lens b. Cornea c. Pupil d. Ciliary muscle

ii. **The underlying criteria for classification can be -**
a. Morphological b. Biochemical c. Ecological d. All of the above

iii. **Which of the following is not an Aldo sugar?**
a. Glucose b. Galactose c. Fructose d. All of the above

iv. **Shape of a human lens can be changed by -**
a. Iris b. Ciliary muscle c. Suspensory muscle d. Ciliary ligament

v. **Engineering requires the use of -**
a. Machines b. Engines c. Structures d. All of the above

vi. **Which among the following is ureotelic?**
a. Protozoa b. Mammals c. Insects d. Amphibians

vii. **A localised group of organisms belonging to the same species constitute a -**
a. Population b. Community c. Ecosystem d. Biome

Group B
(Attempt any 4 questions from the following)

2. **Name the model organism from the Prokaryotic world and give valid reasons for it to be a model organism.** **[1+4]**
3. **Name the Plant species which is highly significant for botanical researches. Enlist its traits that make it very desirable for laboratory study.** **[1+4]**
4. **Enlist five points of difference between Bird flying and Aircraft.** **[5]**
5. **What is Anomeric carbon? What do you mean by α-anomer and β-anomer? Draw the structure of α-Glucose.** **[1+2+2]**
6. **Enlist 5 points of difference between Prokaryotic and Eukaryotic cell.** **[5]**
7. **Draw the Fischer and Haworth structure of Galactose.** **[5]**

--- END OF QUESTION PAPER ---

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ACADEMY OF TECHNOLOGY

Internal Examination (CA3), Even Semester, 2026

Programme/Discipline: CSE/CSBS | **Semester:** 4th

Paper Name: Environmental Science | **Paper Code:** MC-401 | **UPID:** 4421

Date of Examination: 13/05/2026 | **Duration:** 60 min. | **Full Marks:** 25

Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Question No.	1(i)	1(ii)	1(iii)	1(iv)	1(v)	1(vi)	1(vii)	2	3	4	5	6	7
CO Mapping	CO3	CO3	CO2	CO2	CO1	CO1	CO2	CO2	CO3	CO3	CO2	CO1	CO2
Bloom's Level	1	1	2	3	1	1	2	3	2	3	2	2	3

Group A (Attempt any five questions from the following) [1 × 5 = 5]

1. (i) Which one of the following is not considered as adverse effect of global warming?
(a) Melting of ice caps.
(b) Increase in CO_2 concentration in atmosphere.
(c) Decrease in DO content of water reservoir.
(d) Increase in water level height in coastal areas.

(ii) Which one of the following gas does not contribute to the global warming?
(a) Methane, (b) acetylene (c) sulphur dioxide, (d) carbon dioxide.

(iii) The minimum sound pressure that the human ear can detect is
(a) 10 Pa, (b) 20 μPa, (c) 200 μPa,m (d) 2 μPa.

(iv) In a given area two noise levels, 90 dB and 120 dB are active. Intensity ratio of these noise levels is (a) 100, (b) 1000, (c) 10, (d) 10000.

(v) Energy flow of an ecosystem is
(a) non-directional, (b) unidirectional, (c) cyclic, (d) bi-directional.

(vi) Which one of the following is an example of artificial ecosystem?
(a) Pond (b) National park, (c) Valley between two hills, (d) Reserved Forest.

(vii) Which one of the following is not related to USW?
(a) Measuring depth of ocean. (b) Measuring DO content of water.
(c) Scanning the images of heart. (d) Measuring thickness of coating on metal surface.

Group B (Attempt any four questions from the following) [5 × 4 = 20]

2. What is occupational noise? Give one example.
In a given area the following noise levels are active: 90 dB, 95 dB and 100 dB. Calculate the total loudness in that area.

3. (a) What is albedo? (b) State the selection rules for GHG. Hence explain why nitrogen and HCl are not considered GHG.

4. Show that equivalent surface temperature of any planet (T_e) in $°K$ is given by $T_e = \left[\frac{S(1-\alpha)}{4\sigma}\right]^{1/4}$. Hence show that T_e of Earth is $255°K$. Symbols have usual significances.
Given: $\sigma = 5.63 \times 10^{-8}\ W/m^2.°K^4$, S for Earth is 1372 W/m^2 and Albedo for Earth is 0.3.

6. What is the standard reference value of sound intensity? Hence define zero dB. Explain why increase in noise level by only 3 *dB* is harmful to human health.

6. Explain ecological pyramid with diagram. Classify abiotic components with examples.

7. (a) Define Noise Threshold Limit Value (NTLV).

(b) In a work area the different noise levels are read as 95 dB for 1 hr., 100 dB for 45 min. and 90 dB for 2 hrs. Identify whether the existing condition is healthy.

Given: Noise dose limit values of 90 dB, 95 dB and 100 dB are 8 hrs., 4 hrs. and 2 hrs. respectively.